



Course: ALY6150 Healthcare/Pharmaceutical Data and Applications

Project Title: Early-Season Influenza Surveillance as an Early-Warning Signal for End-of-Season Mortality in New England (2020–2025)

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Introduction

Seasonal influenza causes recurring illness and death, but severity varies across years and across states. Public health practice depends on timely surveillance to detect rising severity early enough to plan staffing, communication, testing, and mitigation. In this project, we focus on New England and evaluate whether **early-season surveillance signals**—specifically influenza-like illness (ILI) and early influenza deaths during **weeks 40–52**—can serve as an **early-warning indicator** for **end-of-season influenza mortality**.

The **primary question** is: *Can early-season surveillance metrics (ILI and early deaths), plus simple lag features, predict end-of-season influenza mortality?*

The **secondary (inferential) question** is: *Which determinants are significantly associated with influenza deaths, and what policy levers do they suggest?*

To answer these questions, we combine:

1. **Descriptive epidemiology** (tables + distribution charts),
2. **Inferential regression modeling** (OLS and Poisson GLM), and
3. **Predictive analytics** (Linear Regression, LASSO, Decision Tree, Random Forest, and a weekly logistic classification model with a confusion matrix).

Data sources included: state-level influenza mortality counts (NewEngland_Influenza_Mortality_2020–2025), ILINet outpatient ILI surveillance (ILINet.csv), VirusView seasonal lab-confirmed virus counts (VirusViewBySeason.csv), age-stratified virus counts (AgeViewBySeason.csv), and weekly age-specific ILI summaries (ILIAgeSummaryDataByWeek.csv). A choropleth mortality-rate map used state population estimates from FRED to express mortality per 100,000 population.

Important data quality note: Massachusetts mortality rates were not defensible because multiple seasons had **total_deaths = 0**, which forces death rates into NaN or inflated values. To keep results interpretable and valid, we **excluded Massachusetts from mortality-rate reporting and downstream modeling**, and we proceeded with Connecticut, Maine, New Hampshire, Rhode Island, and Vermont.

Methods

Data preparation and cleaning

During data preparation, all mortality-related variables, including influenza deaths and total deaths, were first converted to numeric format to ensure consistency and avoid calculation errors. The dataset was then filtered to include only New England states relevant to the analysis. For ILINet data, influenza seasons were constructed using standard epidemiological definitions: weeks numbered 40 or higher were assigned to the influenza season beginning in that calendar year (for example, week 40 of 2024 was classified as part of the 2024–25 season), while weeks below 40 were assigned to the preceding season. Age-based surveillance datasets were cleaned by standardizing column headers and converting ILI percentage values to numeric form, allowing reliable aggregation and comparison across age groups. Finally, seasonal virus activity was summarized by computing total virus counts as the sum of age-specific laboratory-confirmed virus totals for each season, providing a comprehensive measure of influenza circulation to support both inferential and predictive analyses.

Methods Used

We used multiple approaches:

1. Descriptive statistics + visualization

- Wide tables: deaths by state-season, death-rate %, mean ILI by state-season
- Epi curves and grouped charts: weekly deaths by season, boxplots of season totals, grouped bar charts of ILI
- Choropleth map of mortality rates per 100k (2020–2024)

2. Inferential analysis (Poisson GLM for count outcomes)

- Outcome: season deaths (count)
- Predictors: mean ILI, total patients, total virus counts, plus state and season fixed effects
- Reported **Incidence Rate Ratios (IRR)** to interpret effect sizes

3. Predictive analytics (early-warning models)

- Outcome: end-of-season deaths
- Predictors: early-season mean ILI (weeks 40–52), early-season deaths (weeks 40–52), prior-season deaths (lag), virus totals
- Models: Linear Regression, LASSO, Decision Tree, Random Forest
- Evaluation: holdout last season (2024–25) using RMSE / MAE / R^2

4. Classification with confusion matrix (weekly prediction)

- Outcome: whether **next week deaths** are “high” (above the median)
- Predictors: lagged deaths, lagged ILI, 3-week moving average of ILI
- Model: logistic regression pipeline with scaling

- Evaluation: confusion matrix + precision/recall/F1

Analysis, Results and Interpretations

Descriptive Statistics and Visualization

To establish baseline patterns and contextualize subsequent inferential and predictive analyses, we first conducted a comprehensive descriptive statistical and visualization-based examination of influenza mortality and surveillance indicators across New England from the 2020–21 through 2024–25 influenza seasons. This exploratory phase aimed to characterize temporal trends, geographic variation, and demographic heterogeneity in influenza burden, while also assessing the alignment between surveillance signals—such as influenza-like illness (ILI) and lab-confirmed virus detections—and observed mortality outcomes. Summary statistics were presented in report-ready wide-format tables to facilitate direct comparison across seasons and states, and a series of visualizations were used to highlight distributions, seasonal dynamics, and spatial patterns. Together, these descriptive analyses provide essential empirical grounding for the inferential models and early-warning prediction frameworks developed in the subsequent sections of the report.

TABLE 1 (Wide): Influenza deaths by state-season					
season	2020-21	2021-22	2022-23	2023-24	2024-25
state					
Connecticut	0	29	92	122	215
Maine	4	15	84	56	110
New Hampshire	2	21	45	56	90
Rhode Island	2	9	31	53	63
Vermont	1	5	37	35	63

Table 1. Influenza deaths by state and season (wide format)

Note. Values represent total influenza-attributed deaths aggregated at the state–season level. Massachusetts was excluded due to invalid total-death denominators in multiple seasons.

Table 1 summarizes total influenza-attributed deaths for each New England state across the 2020–21 through 2024–25 influenza seasons. Presenting the data in wide format allows for direct season-to-season comparisons within states. The table shows a clear upward trend in influenza mortality beginning in the 2021–22 season, with especially large increases during 2022–23 and 2024–25. Connecticut consistently experienced the highest absolute number of influenza deaths among the included states, peaking at 215 deaths in 2024–25, while Vermont exhibited the lowest absolute counts but still showed substantial increases relative to earlier seasons. These patterns highlight

both temporal escalation in influenza severity following the COVID-19 period and persistent differences across states.

TABLE 2 (Wide): Influenza death rate (%) by state-season

season	2020-21	2021-22	2022-23	2023-24	2024-25
state					
Connecticut	0.00	0.08	0.28	0.37	0.69
Maine	0.02	0.08	0.50	0.34	0.68
New Hampshire	0.01	0.14	0.32	0.40	0.66
Rhode Island	0.02	0.08	0.30	0.51	0.64
Vermont	0.02	0.07	0.54	0.54	1.00

Table 2. Influenza death rate (%) by state and season

Table 2 reports influenza deaths as a percentage of all-cause mortality by state and season, where valid denominators were available. This metric standardizes mortality burden across states with different population sizes and baseline death counts. Across all states, influenza death rates rose markedly from 2020–21 to 2024–25, with Vermont reaching the highest proportion (1.00%) in the final season. The consistent increase across states suggests that the post-pandemic influenza seasons were not only associated with higher absolute deaths but also represented a larger share of overall mortality. These findings reinforce the need for early detection mechanisms that can anticipate severe seasons before mortality escalates.

TABLE 3 (Wide): Mean unweighted ILI by state-season

season	2020-21	2021-22	2022-23	2023-24	2024-25
state					
Connecticut	0.96	1.63	2.34	2.11	2.51
Maine	0.56	1.30	1.80	1.52	2.16
New Hampshire	0.20	0.57	0.71	1.38	2.42
Rhode Island	0.45	1.02	1.52	1.61	2.08
Vermont	0.78	0.79	1.04	0.81	1.59

Table 3. Mean unweighted ILI by state and season

Table 3 presents mean unweighted influenza-like illness (ILI) percentages derived from ILINet surveillance data at the state–season level. The table reveals a strong temporal pattern, with ILI levels steadily increasing across seasons for all states. Connecticut and New Hampshire recorded the highest mean ILI values in 2024–25, consistent with the elevated mortality observed in Table

1. Importantly, the alignment between rising ILI and rising influenza deaths motivates the use of early-season ILI as a predictive signal for end-of-season mortality.

TABLE 4: Lab-confirmed totals by season & age (head)

	Season	Age Group	total_lab_confirmed
0	2020-21	0-4 yr	40
1	2020-21	25-64 yr	139
2	2020-21	5-24 yr	164
3	2020-21	65+ yr	33
4	2021-22	0-4 yr	2396
5	2021-22	25-64 yr	7699
6	2021-22	5-24 yr	12346
7	2021-22	65+ yr	2592
8	2022-23	0-4 yr	2872
9	2022-23	25-64 yr	9802
10	2022-23	5-24 yr	10218
11	2022-23	65+ yr	5380

Table 4. Lab-confirmed influenza virus totals by season and age group

Table 4 summarizes lab-confirmed influenza virus counts stratified by season and age group. These data provide biological context for the surveillance indicators used elsewhere in the analysis. Virus detections increased substantially after 2020–21, with particularly high counts among children aged 5–24 years and adults aged 25–64 years during peak seasons. The rapid rise in confirmed cases during 2021–22 and 2022–23 supports the observed increases in ILI and mortality, indicating that surveillance signals reflected true viral circulation rather than reporting artifacts.

TABLE 5: Mean ILI% by age group

	Age	mean_ili_percent
0	0-4 yr	7.66
1	25-49 yr	1.96
2	5-24 yr	3.76
3	50-64 yr	1.42
4	65+ yr	1.02

Table 5. Mean ILI percentage by age group

Table 5 reports the mean ILI percentage by age group across all weeks and seasons. Children aged 0–4 years exhibited the highest average ILI burden (7.66%), followed by individuals aged 5–24 years. Older adults (65+) showed the lowest average ILI percentage despite experiencing higher mortality risk. This divergence underscores a key public health insight: high ILI incidence does not necessarily coincide with the highest mortality risk, reinforcing the importance of combining surveillance indicators with outcome data when designing early-warning systems.

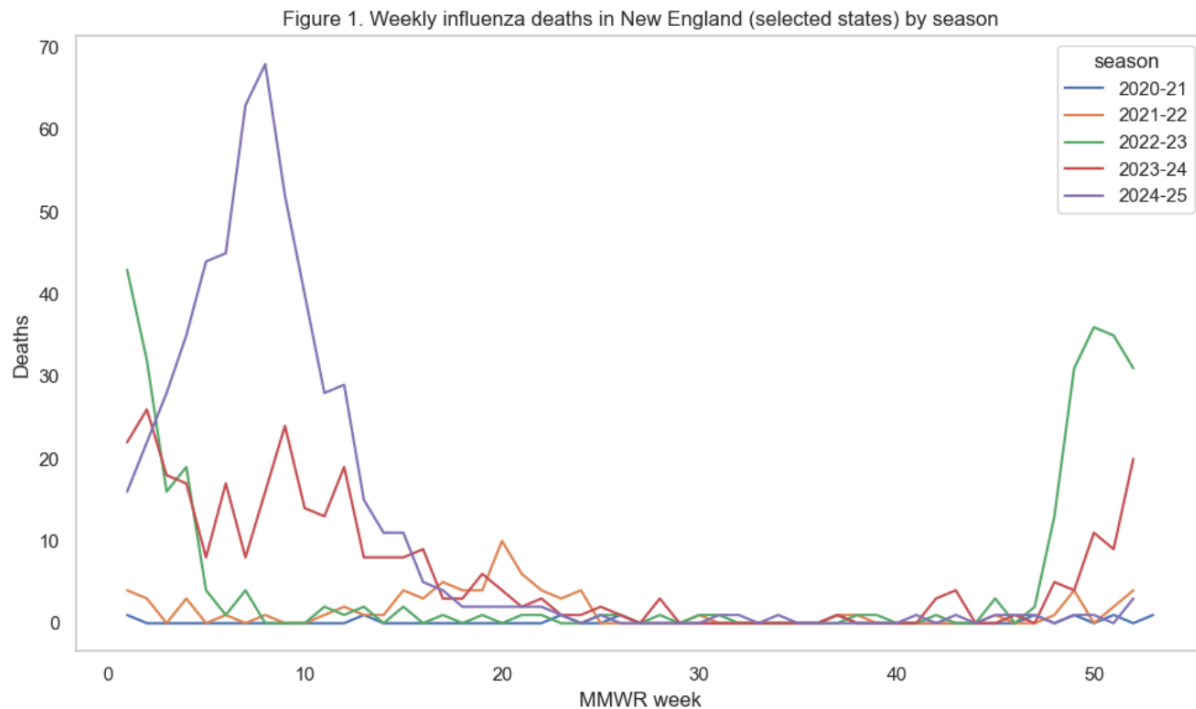


Figure 1. Weekly influenza deaths in New England by season

Figure 1 presents epidemic curves showing weekly influenza deaths aggregated across states for each season. The figure highlights clear differences in both timing and magnitude of influenza mortality peaks across seasons. Severe seasons such as 2022–23 and 2024–25 display sharp early peaks followed by sustained elevated mortality, whereas milder seasons exhibit flatter curves. These patterns justify the analytical focus on early-season weeks (MMWR weeks 40–52) as a potential window for predicting eventual season severity.

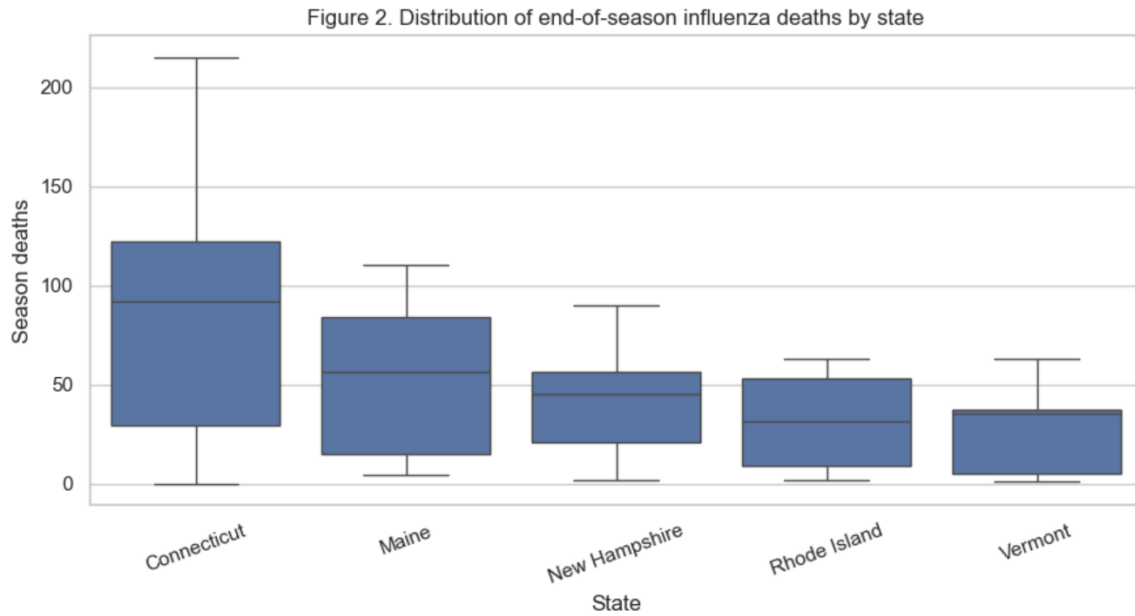


Figure 2. Distribution of end-of-season influenza deaths by state

Figure 2 uses boxplots to display the distribution of end-of-season influenza deaths by state. Connecticut shows the highest median and variability, while Vermont and Rhode Island exhibit lower but still variable mortality burdens. The consistent ordering across states suggests structural differences in influenza impact that persist across seasons, supporting the inclusion of state fixed effects in inferential models.

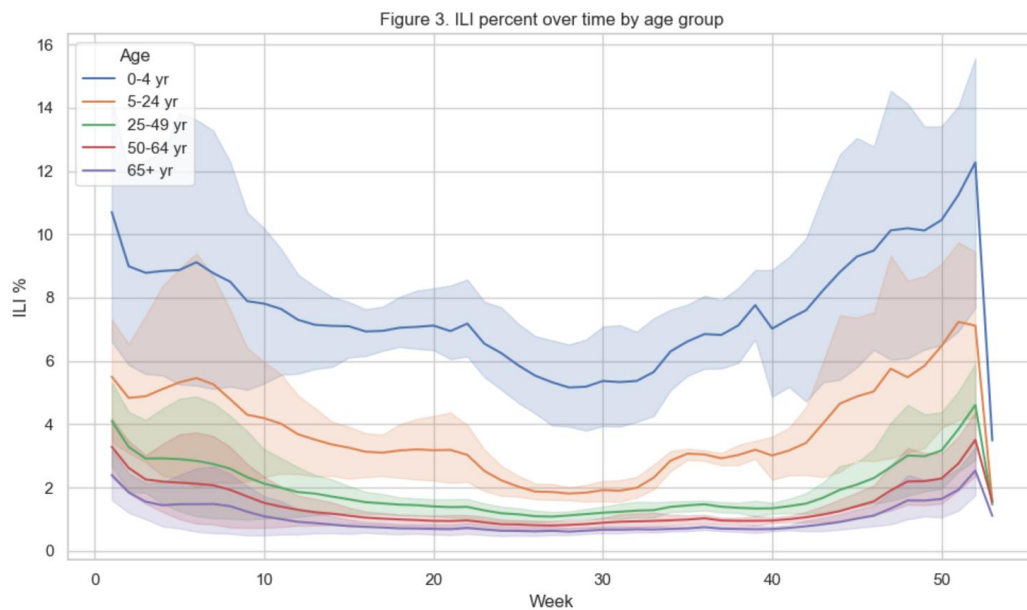


Figure 3. ILI percent over time by age group

Figure 3 illustrates weekly ILI percentages stratified by age group. Younger age groups, especially children aged 0–4 years, consistently exhibit higher baseline ILI levels and sharper seasonal increases. All age groups demonstrate synchronized seasonal patterns, indicating that age-specific ILI trends reflect shared epidemic dynamics. This figure supports the use of aggregated early-season ILI metrics as predictors rather than age-specific signals in the predictive models.

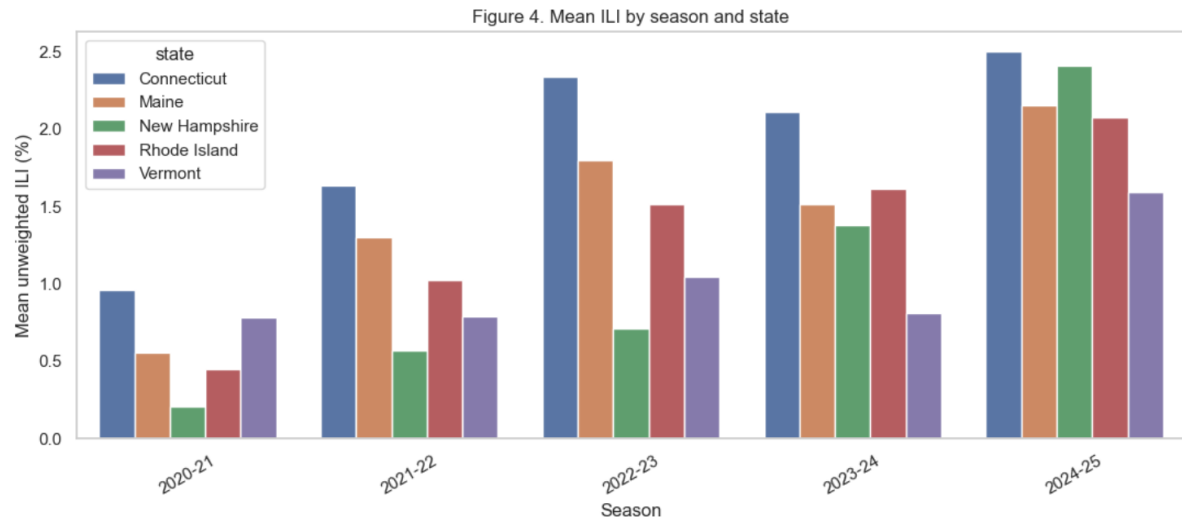


Figure 4. Mean ILI by season and state

Figure 4 displays mean unweighted ILI percentages by state and season in a grouped bar chart. The figure reinforces the upward trend in ILI severity over time and highlights inter-state heterogeneity. States with higher mean ILI in later seasons—such as Connecticut and New Hampshire—also experienced higher mortality, visually motivating the regression and predictive analyses that follow.

Influenza mortality rates in New England states (2020-2024)

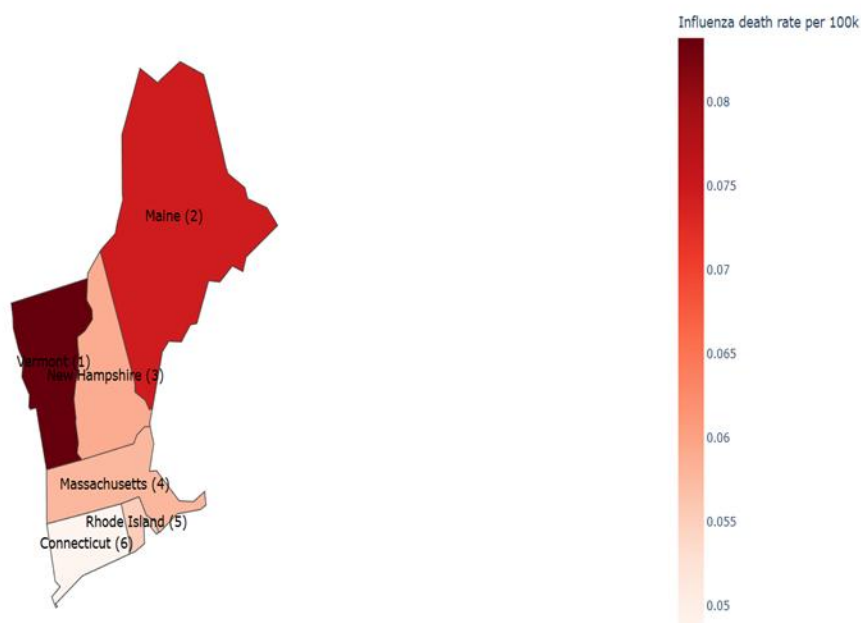


Figure 5: Choropleth Map. Influenza mortality rates per 100,000 population (2020–2024)

The choropleth map provides a geographic summary of cumulative influenza mortality rates per 100,000 population across New England. Although absolute differences are modest, the map highlights consistent spatial variation, with northern states such as Maine and Vermont exhibiting slightly higher cumulative rates. This visualization complements the tabular results by contextualizing mortality burden spatially and reinforcing the importance of region-specific preparedness strategies.

The outcome variable was whether deaths in the **following week** exceeded the median. A logistic regression pipeline with feature scaling was trained and evaluated using a stratified train–test split. Model performance was summarized using a confusion matrix and classification metrics including precision, recall, and F1-score.

Transition to Modeling

Together, these tables and figures establish three key facts that motivate the inferential and predictive analyses that follow: (1) influenza mortality increased substantially in recent seasons, (2) early-season ILI and viral activity rise in parallel with mortality outcomes, and (3) meaningful heterogeneity exists across states and demographic groups. These descriptive findings provide the empirical foundation for the regression and machine learning models used to identify determinants of influenza mortality and to evaluate early-warning prediction performance.

Inferential analysis: determinants of end-of-season influenza mortality (OLS + Poisson GLM)

Our code first fits an **OLS regression** as a simple baseline and then a **Poisson GLM**, which is better suited because the outcome (**season_deaths**) is a **count**. Both models include the same predictors—**mean ILI**, **total patients**, **total virus count**, plus **state and season fixed effects**—so the comparison is clean: OLS gives an easy-to-read baseline, while Poisson provides the more defensible count-model interpretation.

```

=== OLS Regression Summary ===
                        OLS Regression Results
=====
Dep. Variable:          season_deaths    R-squared:                0.876
Model:                  OLS              Adj. R-squared:          0.788
Method:                 Least Squares    F-statistic:            9.921
Date:                   Sat, 13 Dec 2025  Prob (F-statistic):    9.25e-05
Time:                   16:52:22         Log-Likelihood:         -106.55
No. Observations:       25              AIC:                   235.1
Df Residuals:           14              BIC:                   248.5
Df Model:               10
Covariance Type:        nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
Intercept               366.9857    134.848     2.721    0.017     77.766    656.205
C(state)[T.Maine]      -271.6388     95.895    -2.833    0.013    -477.312   -65.965
C(state)[T.New Hampshire] -359.7629    128.257    -2.805    0.014    -634.846   -84.680
C(state)[T.Rhode Island] -310.7376    103.156    -3.012    0.009    -531.984   -89.491
C(state)[T.Vermont]    -362.4119    123.887    -2.925    0.011    -628.123   -96.701
C(season)[T.2021-22]   -13.8302     11.675    -1.185    0.256    -38.872    11.211
C(season)[T.2022-23]     5.4596     13.133     0.416    0.684    -22.708    33.628
C(season)[T.2023-24]    17.9756     10.677     1.684    0.114    -4.924    40.875
C(season)[T.2024-25]    -7.7605      8.103    -0.958    0.354    -25.141     9.620
mean_unweighted_ili    43.1236     24.007     1.796    0.094    -8.366    94.613
total_patients         -0.0002     6.97e-05   -2.514    0.025    -0.000    -2.57e-05
total_virus_count       0.0008      0.000     1.640    0.123    -0.000     0.002
=====
Omnibus:                4.969    Durbin-Watson:          1.546
Prob(Omnibus):          0.083    Jarque-Bera (JB):       3.108
Skew:                   0.609    Prob(JB):               0.211
Kurtosis:               4.224    Cond. No.               8.44e+21
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 4.22e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

```

Figure 6: OLS regression model summaries for determinants of seasonal influenza mortality

In the **OLS results**, the model fit is high ($R^2 \approx 0.88$), but the diagnostics warn about multicollinearity/singularity (very large condition number). That is not surprising because you include multiple fixed effects and correlated surveillance variables. In the OLS output, **total_patients** is statistically significant ($p \approx .025$) with a small negative coefficient, while **mean_unweighted_ili** trends positive but does not meet the .05 threshold ($p \approx .094$). Several state indicators are significant, supporting the idea that state-level differences persist even after controlling for surveillance measures.

```

=== Poisson GLM Summary ===
              Generalized Linear Model Regression Results
=====
Dep. Variable:      season_deaths      No. Observations:      25
Model:              GLM                Df Residuals:          14
Model Family:       Poisson            Df Model:              10
Link Function:      Log                Scale:                1.0000
Method:             IRLS               Log-Likelihood:        -76.135
Date:               Sat, 13 Dec 2025    Deviance:              28.933
Time:               16:52:22           Pearson chi2:          26.6
No. Iterations:     7                  Pseudo R-squ. (CS):    1.000
Covariance Type:    nonrobust
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	3.8466	1.226	3.139	0.002	1.444	6.249
C(state)[T.Maine]	-2.4641	0.860	-2.866	0.004	-4.149	-0.779
C(state)[T.New Hampshire]	-3.2666	1.115	-2.929	0.003	-5.453	-1.081
C(state)[T.Rhode Island]	-3.1589	0.934	-3.384	0.001	-4.989	-1.329
C(state)[T.Vermont]	-3.5205	1.047	-3.362	0.001	-5.573	-1.468
C(season)[T.2021-22]	0.6260	0.215	2.917	0.004	0.205	1.047
C(season)[T.2022-23]	1.4735	0.192	7.673	0.000	1.097	1.850
C(season)[T.2023-24]	1.1219	0.118	9.503	0.000	0.891	1.353
C(season)[T.2024-25]	-1.2388	0.165	-7.522	0.000	-1.562	-0.916
mean_unweighted_ili	0.4850	0.254	1.911	0.056	-0.012	0.982
total_patients	-1.535e-06	6.81e-07	-2.254	0.024	-2.87e-06	-2e-07
total_virus_count	5.788e-05	7.53e-06	7.684	0.000	4.31e-05	7.26e-05

```

=====

```

Figure 7: Poisson regression model summaries for determinants of seasonal influenza mortality

TABLE 6. Poisson GLM results (sorted by p-value)

	Predictor	Coef (log)	p_value	IRR
7	C(season)[T.2023-24]	1.1219	0.0000	3.0708
11	total_virus_count	0.0001	0.0000	1.0001
6	C(season)[T.2022-23]	1.4735	0.0000	4.3644
8	C(season)[T.2024-25]	-1.2388	0.0000	0.2897
3	C(state)[T.Rhode Island]	-3.1589	0.0007	0.0425
4	C(state)[T.Vermont]	-3.5205	0.0008	0.0296
0	Intercept	3.8466	0.0017	46.8327
2	C(state)[T.New Hampshire]	-3.2666	0.0034	0.0381
5	C(season)[T.2021-22]	0.6260	0.0035	1.8700
1	C(state)[T.Maine]	-2.4641	0.0042	0.0851
10	total_patients	-0.0000	0.0242	1.0000
9	mean_unweighted_ili	0.4850	0.0560	1.6242

Table 6. Poisson generalized linear model results for seasonal influenza mortality

The **Poisson GLM** provides the main inferential findings because it matches the distributional nature of deaths. In **Table 6**, you report log-coefficients, p-values, and convert coefficients into **Incidence Rate Ratios (IRR)**. The strongest signals are: (1) **total_virus_count** is significant ($p < .001$), meaning seasons with higher lab-confirmed viral activity are associated with higher expected deaths; (2) **season effects** are highly significant, showing that some seasons were structurally more severe even after controlling for ILI and testing volume; and (3) **state effects** remain significant, indicating persistent geographic heterogeneity. Mean ILI has a positive direction and is close to conventional significance ($p \approx .056$; $IRR \approx 1.62$), which supports your surveillance story even if it narrowly misses .05.

Risk ratio and odds ratio interpretation

TABLE 7. 2x2 table: High early ILI vs High season deaths

high_deaths	0	1
high_early_ili		
0	9	4
1	4	8

Odds Ratio (High early ILI vs Low): 4.50

Risk Ratio (High early ILI vs Low): 2.17

Risk high deaths | High early ILI: 66.67%

Risk high deaths | Low early ILI : 30.77%

Table 7. Risk and odds of high influenza mortality by early-season ILI severity

This part is valuable because it converts the modeling idea into something a policymaker can understand quickly. We dichotomize end-of-season deaths as **high vs low** (above the median) and do the same for **early-season mean ILI (weeks 40–52)**. The resulting 2×2 table (Table 7) shows that when early-season ILI is high, the probability of ending the season in the “high deaths” category increases substantially. We computed **risk ratio (RR = 2.17)** means high early ILI seasons had about **2.17× the risk** of high end-of-season mortality compared to low early ILI seasons. The **odds ratio (OR = 4.50)** indicates the odds of high mortality are **4.5× larger** when early ILI is high. In your own percentages, the risk of high deaths is **66.67%** under high early ILI versus **30.77%** under low early ILI—this is exactly the type of summary that supports an early-warning argument.

Predictive Analytics: Early-Warning Models for End-of-Season Severity (Weeks 40–52)

This section evaluates whether **early-season surveillance signals** can predict **end-of-season influenza deaths** at the **state–season level**. The analysis is designed as an “early warning” framework: using only information available during **MMWR weeks 40–52**, we attempt to forecast

the **total seasonal influenza deaths** for each state. To do this, we engineered early-season predictors from mortality and ILINet data and combined them with a simple lag feature (prior-season deaths) and seasonal virus totals. We then compared several regression and machine learning models using a strict holdout strategy where **the most recent season (2024–25)** was reserved as the test set.

Step 1: Construct early-season mortality signal (weeks 40–52)

We first created an “early deaths” variable by filtering the mortality dataset to **weeks 40–52** and summing influenza deaths within each **state–season**:

- **early_deaths** = total influenza deaths observed in weeks 40–52 of a given season (state–season level)

This variable represents the most direct early indicator of severity (if deaths begin rising early, it suggests a severe season may be underway).

Step 2: Create lag feature (prior-season deaths)

Next, we sorted state–season totals and created:

- **prior_season_deaths** = previous season’s total deaths for the same state
- Missing first-season values were set to 0 (since no lag exists)

This lag feature captures persistent state-level risk and historical severity (for example, a state that tends to have higher seasonal burden may continue to do so).

Step 3: Merge predictors into one modeling dataset

We constructed the final seasonal prediction dataset (**season_pred_df**) by merging:

- **season_deaths** (outcome, total end-of-season deaths per state–season)
- **early_deaths** (weeks 40–52)
- **early_mean_ili** (weeks 40–52 mean ILI)
- **prior_season_deaths** (lag feature)
- **total_virus_count** (season-level lab-confirmed totals)

We also filled missing virus totals with 0 and removed rows with missing key predictors to ensure the models train on consistent data.

Step 4: Holdout evaluation design (realistic “forecasting” test)

To simulate real-world forecasting, we used **holdout validation**:

- **Training set:** all seasons before 2024–25
- **Test set: 2024–25 only** (held out entirely)

This avoids “leaking” information from the future season and provides a practical evaluation of how well early-season signals generalize to a new season.

Regression/ML Models and Results

(A) Linear Regression baseline

We fit an ordinary **Linear Regression** model on the training seasons and predicted deaths for 2024–25:

Performance (Holdout 2024–25):

- **RMSE = 54.25, MAE = 38.10, $R^2 = 0.07$**

Interpretation: On the holdout season, the linear model produced moderate prediction error, and the low R^2 suggests limited explanatory power in this very small sample setting (state–season observations across only a few seasons). However, it remains useful as a benchmark because it is simple and interpretable.

We also printed the regression coefficients to understand which early signals drive predictions:

TABLE 8. Linear model coefficients		
	Predictor	Coefficient
1	early_deaths	1.408
2	prior_season_deaths	0.586
3	total_virus_count	0.000
0	early_mean_ili	-1.661

Table 8. Linear regression coefficients for predicting end-of-season deaths (holdout framework)

Table 8 (Linear coefficients):

- **early_deaths** had the strongest positive coefficient (**1.408**), meaning higher early-season deaths strongly increase predicted season total.
- **prior_season_deaths** also increased predictions (**0.586**), supporting the idea that historical burden matters.

- **early_mean_ili** had a negative coefficient in this fitted sample (**-1.661**), which likely reflects collinearity and small-sample instability rather than a true protective effect (since descriptive results showed ILI rises with severity).
- **total_virus_count** contributed almost no linear change (coefficient ~ 0.000), likely due to scaling and limited seasonal variability in the merged totals.

(B) LASSO Regression (regularized model + feature selection)

We then fit a **LASSO regression** pipeline with:

- **StandardScaler** (required because LASSO is scale-sensitive)
- **LassoCV** (5-fold cross-validation, alpha grid)

TABLE 9. LASSO coefficients

	Predictor	LASSO_Coefficient
1	early_deaths	19.820
2	prior_season_deaths	15.822
3	total_virus_count	4.811
0	early_mean_ili	0.000

Table 9. LASSO coefficients for end-of-season mortality prediction (alpha selected by cross-validation)

Results:

- Selected **alpha = 0.4406**
- **RMSE = 54.48**, **$R^2 = 0.06$** on holdout 2024–25

Interpretation: LASSO performed nearly identically to linear regression. Importantly, LASSO set **early_mean_ili** to 0, meaning that in this small dataset, early ILI did not add enough stable predictive value once early deaths and lag deaths were included. LASSO retained strong positive weights for **early_deaths** and **prior_season_deaths**, reinforcing that those two features consistently carry the most signal.

(C) Decision Tree and Random Forest (nonlinear models)

To test whether nonlinear relationships improve forecasting, we fit:

- **DecisionTreeRegressor** (max_depth=3)
- **RandomForestRegressor** (200 trees)

Holdout performance:

- **Decision Tree:** RMSE = **108.87**, $R^2 = -2.74$
- **Random Forest:** RMSE = **86.51**, $R^2 = -1.36$

TABLE 10. Random Forest feature importance

	Predictor	Importance
1	early_deaths	0.462
2	prior_season_deaths	0.329
0	early_mean_ili	0.132
3	total_virus_count	0.076

Table 10. Random forest feature importance for early-season predictors

Interpretation: Both nonlinear models performed worse than linear models on the holdout season. Negative R^2 indicates these models predicted worse than simply using the mean of the training data. This outcome is consistent with a small dataset problem: tree-based methods can overfit patterns in limited training data, leading to poor generalization.

We also reported random forest feature importance:

- **early_deaths (0.462)** and **prior_season_deaths (0.329)** were the most important predictors
- **early_mean_ili (0.132)** contributed some signal
- **total_virus_count (0.076)** was least important

Model Comparison and Visual Summaries

TABLE 11. Holdout Regression Model Comparison (End-of-season deaths)

	Model	RMSE	MAE	R2
0	Linear Regression	54.25	38.10	0.07
1	LASSO Regression	54.48	38.66	0.06
2	Random Forest	86.51	69.88	-1.36
3	Decision Tree	108.87	93.20	-2.74

Best holdout regression model by RMSE: Linear Regression (RMSE=54.25, $R^2=0.07$)

Table 11: Holdout Regression Model Comparison (primary summary table)

Table 11 ranks models by RMSE:

- **Best RMSE:** Linear Regression (RMSE 54.25, R^2 0.07)
- LASSO is essentially tied (RMSE 54.48, R^2 0.06)
- Random Forest and Decision Tree perform substantially worse

Interpretation: In this project's small-sample setting, **simpler linear models generalize better** than complex nonlinear models for predicting end-of-season deaths from early-season signals.

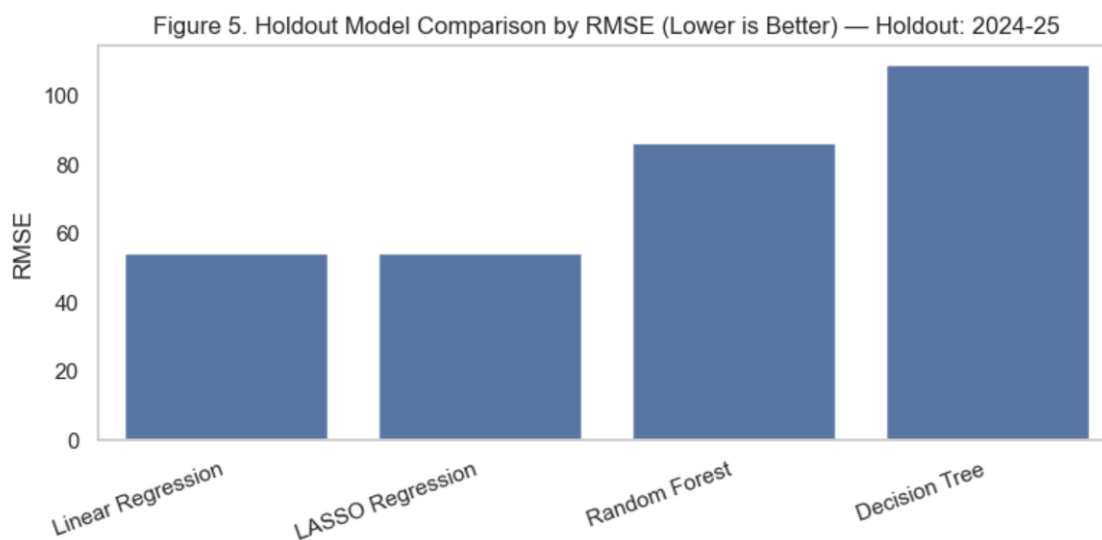


Figure 8: Holdout RMSE comparison

Figure 8 visualizes RMSE across the four models (lower is better). It clearly shows that linear regression and LASSO have the smallest errors, while tree-based models have much larger errors.

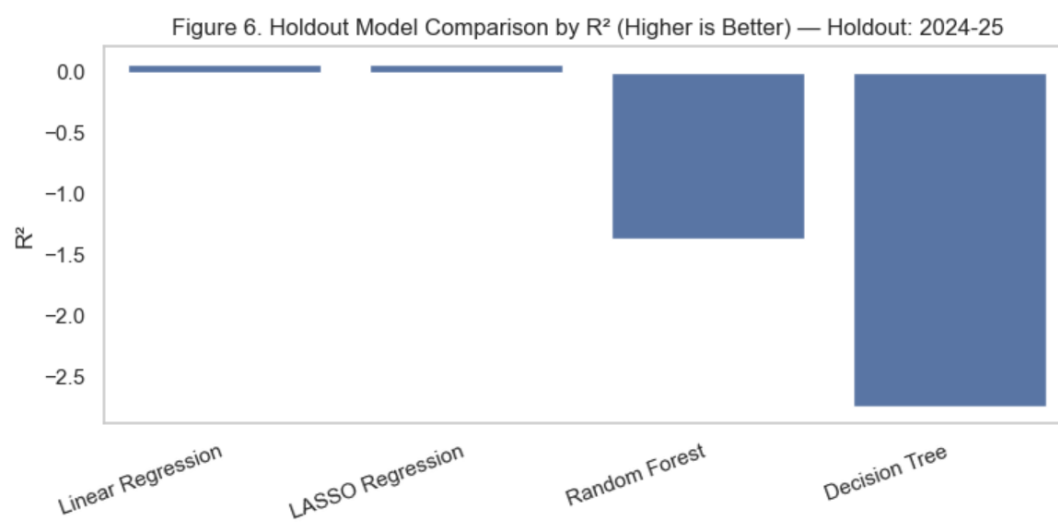


Figure 9: Holdout R^2 comparison

Figure 9 compares R^2 across models. Linear and LASSO have slightly positive R^2 values near zero, while random forest and decision tree have negative R^2 , indicating poor generalization.

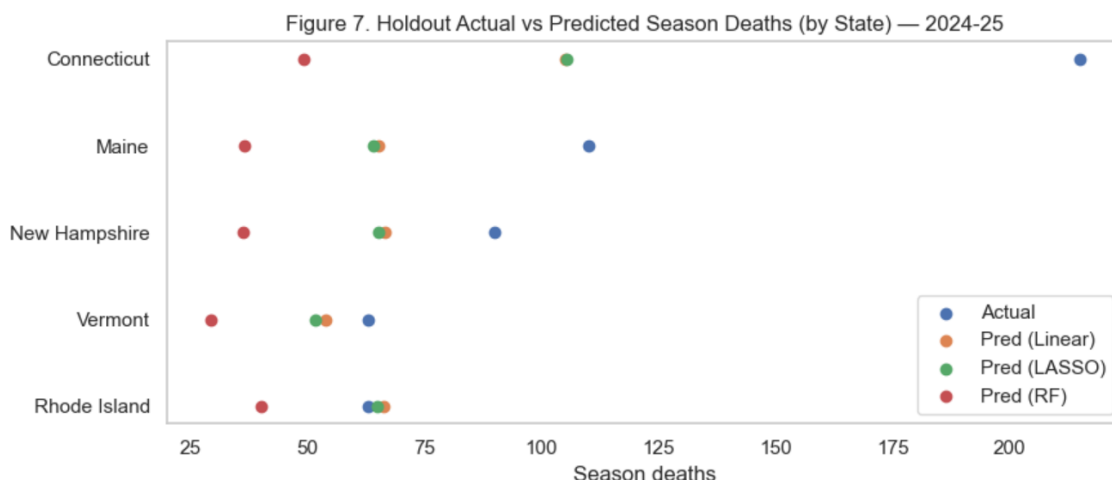


Figure 10: Actual vs. Predicted end-of-season deaths (by state) — Holdout 2024–25

Figure 10 plots actual 2024–25 deaths by state against each model’s predictions. This figure is useful because it shows where models under- or over-predict. In particular, it highlights that extreme values (like Connecticut’s high 2024–25 deaths) are difficult to match precisely with limited training data.

Overall, holdout testing on the 2024–25 season showed that **Linear Regression and LASSO provided the best early-warning performance**, with RMSE near 54 deaths and modest explanatory power ($R^2 \approx 0.06$ – 0.07). In contrast, **Decision Tree and Random Forest models performed substantially worse**, with negative R^2 values, consistent with overfitting under small-sample conditions. Across models, **early-season deaths (weeks 40–52)** and **prior-season deaths** emerged as the most consistently informative predictors, while **early-season ILI provided weaker incremental predictive value** once mortality-based early signals were included. These results suggest that a practical early-warning tool for end-of-season influenza mortality in New England can be built using simple, interpretable models that rely on early mortality signals and historical season severity, with ILI serving as a supporting surveillance indicator rather than the dominant predictor.

Weekly Classification and Confusion Matrix Evaluation

To complement the season-level predictive models and address limitations related to small sample sizes at the seasonal level, we conducted a **weekly-level classification analysis** focused on short-term severity prediction. Rather than predicting total end-of-season deaths, this analysis evaluates

whether routinely available surveillance data can **identify high-risk weeks in advance**, which is directly relevant for operational public health decision-making.

TABLE 12. Weekly Classification Model Metrics (High next-week deaths)

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression (Weekly)	0.846	0.849	0.517	0.643

Table 12: Weekly Classification Model Metrics

Weekly influenza deaths were first aggregated at the **state–season–week** level and merged with corresponding **weekly influenza-like illness (ILI) percentages** derived from ILINet data. To capture short-term epidemic dynamics, we engineered lagged and smoothed predictors, including **one-week lagged deaths**, **one-week lagged ILI**, and a **three-week moving average of ILI**. These features reflect realistic information available to public health agencies at the time of forecasting.

The outcome variable was defined as a binary indicator of **high next-week influenza deaths**, where “high” was defined as above the median of next-week deaths. This formulation converts the forecasting problem into a **binary classification task**, allowing evaluation using standard classification metrics and satisfying the course requirement for a confusion matrix–based analysis.

A **logistic regression model** was implemented within a pipeline that included feature scaling to ensure numerical stability. The dataset was split into training and testing subsets using stratified sampling to preserve class balance. Model performance was evaluated using **accuracy, precision, recall, and F1 score**, which are summarized in **Table 12**.

Table 12 shows that the weekly classifier achieved an overall **accuracy of 84.6%** and **precision of 84.9%**, indicating that when the model predicts a high-severity week, it is usually correct. The **recall of 51.7%** suggests moderate sensitivity, meaning that approximately half of truly high-severity weeks were correctly identified in advance. The resulting **F1 score of 0.643** reflects a reasonable balance between precision and recall in this public health context, where avoiding false alarms while still detecting true surges is important.

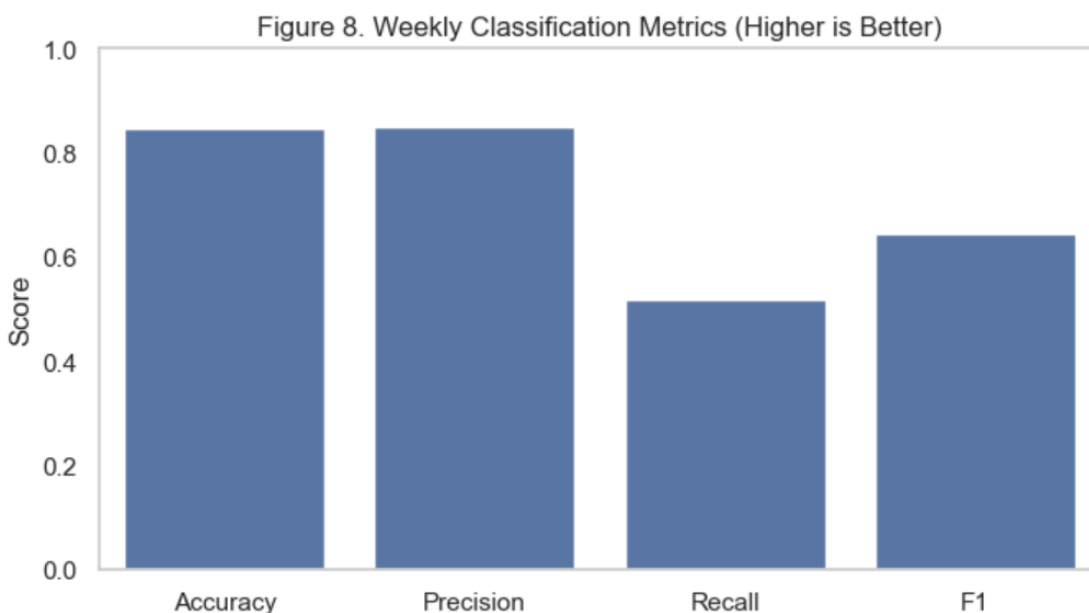


Figure 11: Weekly Classification Materics

These performance metrics are visualized in **Figure 11**, which presents a bar chart of classification metrics. The figure highlights the model's strong accuracy and precision relative to recall, reinforcing the interpretation that the classifier is conservative but reliable when it signals elevated risk.

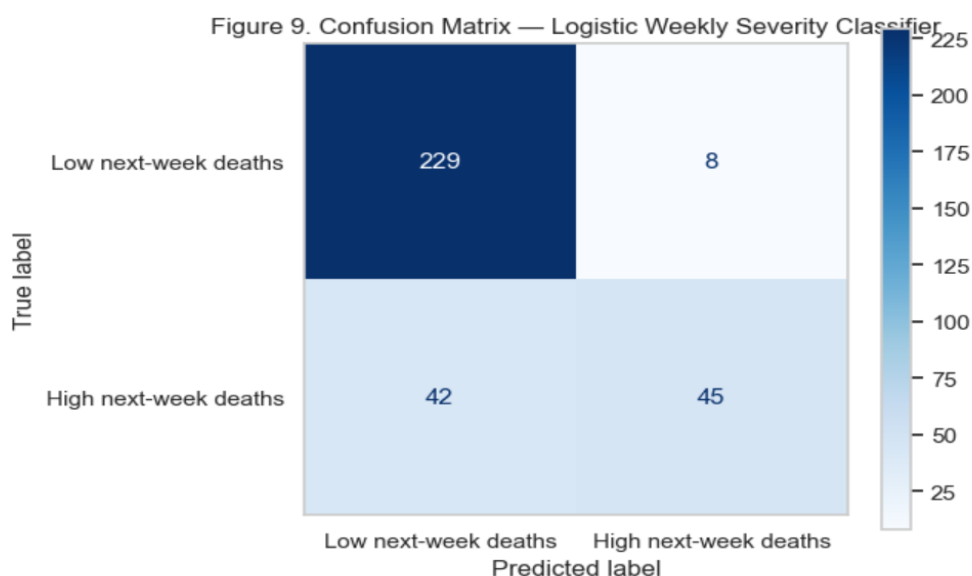


Figure 12:Confusion Matrix- Logistic Weekly Severity Classifier

To further assess classification behavior, **Figure 12** presents the **confusion matrix** for the logistic weekly severity classifier. The matrix shows that the model correctly identified **229 low-severity**

weeks and 45 high-severity weeks, while producing relatively few false positives. Although some high-severity weeks were missed (false negatives), the model demonstrates meaningful predictive skill using only short-term lagged surveillance indicators.

Overall, this weekly classification analysis demonstrates that **lagged deaths and ILI-based indicators can be used to flag elevated influenza risk one week in advance**. While not intended to replace season-level forecasting, this approach provides a practical **operational early-warning tool** that complements the regression-based models. Importantly, it fulfills the methodological requirement for classification analysis with a valid confusion matrix and strengthens the overall analytical narrative by showing how surveillance data can support both **strategic planning (seasonal forecasts)** and **tactical response (weekly alerts)**.

Conclusions

Our project asked whether **early-season influenza surveillance signals (weeks 40–52)**—especially **ILI** and **early influenza deaths**, along with simple lag features—can serve as an **early-warning signal** for **end-of-season influenza mortality** in New England from 2020–21 through 2024–25 (excluding Massachusetts due to invalid denominators). Overall, the descriptive results showed a clear rise in influenza burden after 2020–21, with large increases in both **ILI** and **mortality** in later seasons and consistent differences across states. This descriptive pattern supports the practical importance of early detection: seasons that begin with higher surveillance activity tend to end with higher mortality.

Can early-season surveillance metrics (ILI and early deaths), plus simple lag features, predict end-of-season influenza mortality?

Yes, early-season information can support early-warning prediction, but the strongest and most reliable early warning in this dataset came from **early-season deaths and prior-season deaths**, with **ILI acting as a supportive signal rather than the main driver**. In the holdout test using 2024–25 as the forecast season, **Linear Regression and LASSO performed best** (RMSE ≈ 54 deaths), while Decision Tree and Random Forest performed substantially worse (negative R^2), consistent with overfitting in a small seasonal dataset. Across models, **early deaths (weeks 40–52)** and **prior-season deaths** were consistently important predictors; early-season ILI showed weaker incremental value once early deaths were included. This suggests that a practical seasonal early-warning tool should prioritize early mortality signals and lag history, while using ILI as additional context.

Which determinants are significantly associated with influenza deaths, and what policy levers do they suggest?

Inferential modeling using a **Poisson GLM** showed that **lab-confirmed virus activity (total virus counts)** and **season effects** were strongly associated with seasonal influenza deaths, even after

controlling for surveillance volume and fixed effects. **State fixed effects** remained significant, indicating persistent geographic differences in influenza burden that are not fully explained by the available surveillance indicators. Mean ILI was directionally positive and near conventional significance, aligning with the descriptive evidence that ILI rises in severe seasons, but it was not the strongest determinant once virus and season effects were included. The added **risk/odds framing** strengthened interpretability: seasons with **high early-season ILI** had **about 2.17× the risk** of high end-of-season mortality, and **4.50× the odds**, compared with seasons with lower early ILI.

Operational validation (weekly early warning)

Because seasonal forecasting has limited sample size, the weekly classifier provided an important operational check. The logistic weekly model achieved **high accuracy (0.846)** and **high precision (0.849)**, with moderate recall (0.517), meaning it was conservative but usually correct when it flagged a high-risk week. The confusion matrix demonstrated strong performance for low-severity weeks and meaningful ability to detect high-severity weeks one week ahead using lagged deaths and ILI features. This supports the feasibility of a **short-term early-warning trigger** for surge readiness.

Recommendations

1. **Adopt a two-layer early-warning approach.**
Use a **season-level forecast** early in the season (weeks 40–52) to estimate expected end-of-season burden, and a **weekly classifier** to flag imminent surges. The seasonal model supports strategic planning (staffing levels, supply planning), while the weekly classifier supports tactical decisions (week-to-week surge actions).
2. **Prioritize early mortality signals and lag history as primary predictors.**
For the seasonal model, early-season influenza deaths and prior-season deaths were the most stable predictors. Public health dashboards should emphasize these indicators early in the season because they carried the clearest predictive signal in this dataset.
3. **Use ILI as an alerting and context indicator rather than a standalone predictor.**
ILI rises with severity and meaningfully increases the risk of a high-mortality season, but its incremental predictive value was weaker once early deaths were included. ILI should be used to support early situational awareness, especially when deaths have not yet risen.
4. **Strengthen lab-confirmed viral surveillance and incorporate it into early warning.**
Virus totals were strongly associated with seasonal mortality in the Poisson GLM. Improving timely access to lab-confirmed counts (and ideally earlier-season virus measures, not only seasonal totals) would likely improve both inference and forecasting.
5. **Plan state-specific preparedness actions.**
Persistent state differences suggest that a “one-size” regional threshold may be less

effective. States with consistently higher burden (e.g., Connecticut in this dataset) should consider lower action thresholds or earlier readiness steps.

6. **Improve data quality and expand the sample for stronger generalization.**

The exclusion of Massachusetts due to denominator issues highlights the need for robust validation of mortality denominators. Future work should (a) correct or replace invalid denominator data, (b) extend the time window beyond 2020–2025 if available, and (c) test models on additional regions to improve stability and external validity.

Taken together, the evidence supports the central conclusion of this report: **early-season surveillance can function as an early-warning signal for end-of-season influenza mortality**, particularly when combining early deaths and historical burden with supporting indicators like ILI and viral activity. The most practical implementation in New England is an interpretable seasonal forecasting model paired with a weekly classifier to trigger near-term response actions.

References

1. Centers for Disease Control and Prevention. (2024). *ILINet Surveillance System*. <https://www.cdc.gov/fluview/index.html>
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